When and Why Vision-Language Models Behave like Bags-Of-Words, and What to Do About It?

Victor Guichard
ENS Paris-Saclay
guichardvictor@gmail.com

Eliott Vigier
ENS Paris-Saclay
eliott.vigier@gmail.com

Paul Chu
ENS Paris-Saclay
paulchuparis@gmail.com

1. Introduction

In recent years, vision-language models (VLMs) such as CLIP [4] or BLIP [5] have attracted interest for their ability to learn joint representations of images and text using contrastive learning. CLIP [4] is designed to align visual and textual modalities by training on large-scale datasets, enabling robust zero-shot performance on various tasks. BLIP [5] leverages image and language features to achieve results on vision-language tasks. Contrary to what we stated in our proposal, we did not reduce the datasets and used the full datasets instead.

Despite their capabilities, these models often behave like “bags-of-words,” meaning they can overlook the compositional structure of language like word order and the nuanced relationships between objects and their attributes.

NegCLIP [8] was introduced to overcome this limitation by using a composition-aware training strategy. Unlike CLIP [4] training, NegCLIP [8] utilizes hard negatives generated from perturbed captions (shuffled verb and nouns for instance) and selected alternative images during training. Therefore, the model pay closer attention to the order of words and the relationships between objects.

To evaluate the compositional understanding of VLMs, we use the ARO benchmark. The ARO benchmark comprises several datasets: Visual Genome is used for testing object relations and attributes, COCO and Flickr30k for assessing sensitivity to word order.

In our study, we will reproduce the major results reported in the research paper for BLIP [5], CLIP [4], and NegCLIP [8]. Moreover, we propose an extension of the analysis to investigate whether image generative models also exhibit bag-of-words behavior.

2. Reproduction

2.1. COCO training and validation

In the research paper [8], they used the full Common Objects in Context (COCO) 2014 dataset for both training and validation. It representes 82,783 training images and 40,504 validation images. To overcome the bag-of-words behavior, they applied a hard negative mining strategy during finetuning. It involves generating hard negative captions by perturbing the original captions and selecting alternative images as negatives. For example, given the original caption :

> “remarkable scene with a blue ball behind a green chair”

We generated several hard negative captions:

- Shuffle nouns and adjectives: “green ball with a remarkable chair behind a blue scene” Shuffle everything but nouns and adjectives: “remarkable scene behind a blue ball with a green chair”
- Shuffle trigrams: “a green chair remarkable scene with a blue ball behind”
- Shuffle words within each trigram: “scene with remarkable a ball blue a green behind chair”

2.2. VG-R Evaluation

We first reproduce the evaluation of pre trained CLIP [5] and BLIP [4] models on the VG-R Dataset.

Visual Genome (VG) [2] is a large-scale collection of over 100,000 images annotated with objects, attributes, and relationships. The author built two datasets from here:

- Visual Genome Relation (VG-R): This dataset assesses whether a model can correctly determine the order of relational expressions. Given an image and a relation in the form X relation Y, the model must choose between X relation Y and Y relation X. Test cases include various types of relationships, such as prepositional phrases (e.g., ”the dog is behind the tree” vs. ”the tree is behind the dog”) and verb-based relations (e.g., ”the horse is eating the grass” vs. ”the grass is eating the horse”).
- Visual Genome Attribution (VG-A): This dataset evaluates a model’s ability to accurately associate attributes with objects. For example, the model must differentiate between phrases like ”the crouched cat and the open door” and ”the open cat and the crouched door.”

For the evaluation of pre trained models, we decided to only focus on the VG-R as our reproduction already managed to achieve the same performance on this dataset, we decided to move forward on the next steps of our project.

Similarly to the author, we also removed the symmetric words such as 'adjusting', 'attached to', 'between'.

2.3. Reproduction of NegClip

In order to reproduce NegCLIP [8], we finetuned the *ViT-B/32* variant of CLIP [4] on the COCO dataset with hard negatives, as earlier. We used a learning rate of 1×10^{-6} and trained the model for 5 epochs. Due to resource limitations, the batch size was reduced from 256 to 64. We evaluated the performance of the finetuned model on both the *Visual Genome Attribution* and *Visual Genome Relation* datasets. Additionally, we evaluated the model on CIFAR10 to verify that it does not lose performance on the base task. The training was monitored using retrieval metrics such as mean and median ranks, recall at various thresholds. For comparative analysis, we retained two checkpoints: one from epoch 2 and one from epoch 5.

3. Do image generative models behave like bag-of-words?

3.1. Compositional visual generation

Initially, we proposed extending our work by redoing the Visual Genome benchmark using newly generated images with alternative captions. However, during image generation, we observed that many generative models, such as Flux [3] and Stable Diffusion v2.1 [6], struggled to produce images that accurately reflected the given captions, as illustrated in Figures 3. Consequently, we decided to develop a benchmark based on the methodologies and models proposed in [7].

3.2. Our proposed benchmark

The benchmark assesses image generation quality and caption alignment using 1000 samples from the VG-R dataset. We reduced the number of samples due to limited gpu budget and to favor testing a second diffusion model.

For each dataset element:

- We generate a 256x256 image using the default pipeline hyper parameters and an augmented prompt: *"A photo corresponding to the following description: " + wrong caption*. This ensures a more descriptive and informative input for the generative model.
- We apply NegCLIP [8] to identify the closest matching prompt from the dataset.
- We measure accuracy based on correctly retrieved prompts.

This method has limitations: (1) Image generation is still an open challenge, and (2) NegCLIP [8] achieves around 80% accuracy on VG-R. However, it provides a solid starting point for evaluating both image quality and how well the generated image aligns with the captions.

4. Results

4.1. Reproduction of VG-R of CLIP and BLIP

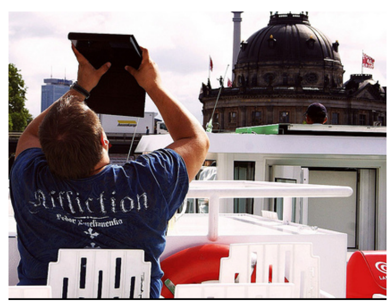
As depicted in our reproduction in table 1, we managed to reproduce almost the exact same results as written on the paper.

Model	Paper	Reproduction
CLIP VIT-B/32	59%	58.37%
BLIP	59%	61.63%

Table 1. Vision-Language models accuracy on VG-R Dataset

We also explored error cases, as illustrated in Figure 1. An interesting insight from our qualitative evaluation is that some original captions appear questionable. For instance, the caption 'the man is behind the building' could be considered inaccurate. While some keywords helped filtering some issues such as "sitting behind" and "standing behind", the "is behind" was not. This observation suggests the presence of potentially noisy samples within the VG-R dataset.

(a) "the printer is to the left of the arm": 0.2927
"the arm is to the left of the printer": 0.2793



(b) "the building is behind the man": 0.226
"the man is behind the building": 0.213

Figure 1. CLIP score on images of VG-R dataset on some miss classified samples

4.2. Reproduction of NegCLIP

We retained two checkpoints (epoch 2 and epoch 5) for comparative analysis. In our evaluation, the model achieved

81% on the Visual Genome Relation task and around 71% on the Visual Genome Attribution task. These values are the same of the paper's reported metrics.

Checkpoint	VG-Relation Accuracy	VG-Attribution Accuracy
Epoch 2	81.18%	71.79%
Epoch 5	80.95%	71.38%
Paper Reported	81%	71%

Table 2. Evaluation on Visual Genome Relation and Attribution tasks.

We observed that, regardless of the dataset, the model checkpoint at epoch 2 slightly outperforms that at epoch 5. This could be explained by the fact that the epoch 2 model shows a higher text-to-image recall during validation.

Metric	Epoch 2	Epoch 5
Image-to-Text Mean Rank	21.02	22.10
Image-to-Text Median Rank	5.0	5.0
Text-to-Image R@10	0.6699	0.6641

Table 3. Comparison of key retrieval metrics between epoch 2 and epoch 5 on evaluation dataset.

Additionally, we evaluated the model in a zero-shot setting on CIFAR-10 to observe if the model does not lose performance on the base task. Our results indicate a zero-shot accuracy of 84.84% at epoch 2 and 83.64% at epoch 5. Here again, the model checkpoint at epoch 2 is better. However, the accuracy is lower than the 94% reported in the paper.

Checkpoint	CIFAR-10
Epoch 2	84.84%
Epoch 5	83.64%
OpenAI CLIP	88.82%
OpenAI CLIP Reported	95%
Paper Reported	94%

Table 4. Zero-Shot accuracy on CIFAR-10.

This pattern is also observed for CLIP (a decrease of 6-7% compared to the reported results). One explanation could be that we did not employ prompt ensembling during the evaluation and only used the following prompt: *"a photo of a(n) [label].*

4.3. Image generative models behaving like bag-of-words

We reported the accuracy of our diffusion models pipeline on our benchmark in table 5

Our results indicate that the models struggle to generate images that closely align with the given captions, especially when the prompt describes unrealistic scenarios, such

Model	Accuracy
Stable Diffusion Turbo XL	7%
stable-diffusion-3.5-large-turbo	4.431%

Table 5. Generative Models accuracy based on NegCLIP similarities

as "the grass eating the horse." One limitation of our experiment was the absence of a secondary metric to quantify the reduction in similarity between the generated image and the original image. Including this metric would have provided deeper insight into whether the model's failure came more from poor image generation or misinterpretation of the caption.

(a) Image for *"the door is to the left of the man"*

(b) Image for *"the shirt is wearing the man"*

Figure 2. Stable Diffusion v3 Large Turbo still has some trouble to generate based on the given prompt.

We have also seen many failure cases of just generating a proper image with our given prompt. Figure 2 shows some of the image generated for some prompt of the VG-R dataset. One way to potentially automatically improve upon this issue could be by using an LLM to generate more rich and more adapated prompt using the caption.

5. Conclusion

In this project, we successfully reproduced the results presented in the paper "When and Why Vision-Language Models Behave like Bags-Of-Words, and What to Do About It?" by evaluating multiple vision-language models on the VG-R dataset and retraining NegCLIP [8] on the COCO dataset. Our main deviation from the original paper was our zero-shot evaluation on CIFAR-10, where our NegCLIP [8] model performed slightly worse than the one proposed in the study.

We then extended this work to text-conditioned generative models, highlighting a critical issue in caption alignment, particularly in challenging scenarios where the generative model fails to produce a correct image. Recent advancements, such as Janus [1] and 4o image generation, are addressing these alignment challenges. Additionally, new datasets like CoBSAT [9] emphasize the alignment of visual and textual information in text-to-image tasks, providing a valuable benchmark for future research.

These developments mark an important step toward improving vision-language consistency improving both multimodal language model and generative models.

References

[1] Xiaokang Chen, Zhiyu Wu, Xingchao Liu, Zizheng Pan, Wen Liu, Zhenda Xie, Xingkai Yu, and Chong Ruan. Janus-pro: Unified multimodal understanding and generation with data and model scaling. *arXiv preprint arXiv:2501.17811*, 2025. 4

[2] Ranjay Krishna, Yuke Zhu, Oliver Groth, Justin Johnson, Kenji Hata, Joshua Kravitz, Stephanie Chen, Yannis Kalantidis, Li-Jia Li, David A. Shamma, Michael S. Bernstein, and Fei-Fei Li. Visual genome: Connecting language and vision using crowdsourced dense image annotations, 2016. 1

[3] Black Forest Labs. Flux. https://github.com/black-forest-labs/flux, 2024. 2

[4] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation, 2022. 1, 2

[5] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision, 2021. 1

[6] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10684–10695, 2022. 2

[7] Mert Yuksekgonul, Federico Bianchi, Pratyusha Kalluri, Dan Jurafsky, and James Zou. When and why vision-language models behave like bags-of-words, and what to do about it?, 2023. 2

[8] Mert Yuksekgonul, Federico Bianchi, Pratyusha Kalluri, Dan Jurafsky, and James Zou. When and why vision-language models behave like bags-of-words, and what to do about it? In *International Conference on Learning Representations*, 2023. 1, 2, 4

[9] Yuchen Zeng, Wonjun Kang, Yicong Chen, Hyung Il Koo, and Kangwook Lee. Can mllms perform text-to-image in-context learning? *arXiv preprint arXiv:2402.01293*, 2024. 4

(a) Flux.1-dev

(b) Stable Diffusion v2.1

Figure 3. Image Generated with prompt: "The grass is eating the horse"